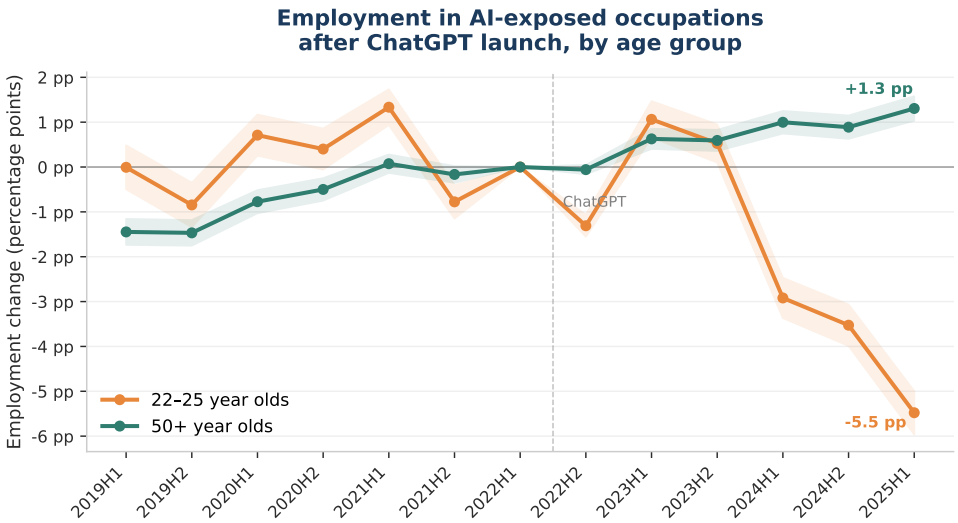
Same Storm, Different Boats: Generative AI and the Age Gradient in Hiring

Magnus Lodefalk[a,b] • Lydia Löthman[a] • Michael Koch[c] • Erik Engberg[a,b]

[a]Örebro University [b]RATIO Institute [c]Aarhus University / Kiel Institut

Key finding: After ChatGPT's launch, employment of **22–25 year olds** in AI-exposed occupations began declining relative to less exposed occupations, reaching **−5.5%** by early 2025, while employment of **workers over 50 rose by 1.3%**. The gap is accelerating.



Note: Employer×quartile and employer×month FE. Ref. period: 2022H1. 95% CI shown.

What we found

- The widely noted decline in job postings reflects **monetary tightening**, not AI. A natural timing test confirms this: the decline begins with the Riksbank rate hike (April 2022), seven months *before* ChatGPT, and there is no correlation between AI exposure and interest-rate sensitivity.
- Beneath this aggregate trend, AI is reshaping **who gets hired**. Within the same firms, young workers in AI-exposed occupations (e.g., software developers, customer service agents, administrative assistants) are disproportionately losing employment, a shift invisible in aggregate statistics.
- The effect **accelerates over time**: by early 2025, 22–25 year olds in high-AI occupations had lost **5.5%** of employment relative to low-AI occupations, up from near zero at launch. Workers aged 26–30 follow a similar trajectory (−4.9% by 2025H1). Meanwhile, workers over 50 *gained* 1.3%. The result mirrors US findings (Brynjolfsson et al., 2025) and contrasts with a Finnish null (Kauhanen, 2025); our within-employer design using **full-population** data may capture composition shifts not visible at the occupation level.
- The effect is **stronger for women**: the average treatment effect for young women is −1.6%, more than double the −0.7% for young men.

Policy implications

- **AI operates on employment composition**, not only aggregate demand. Targeted interventions are needed alongside broader economic policy.
- Candidate responses include **entry-level transition pathways**: vocational education (YH) and apprenticeships integrating AI tools, though their effectiveness for this specific challenge remains to be tested.
- **The silver lining**: evidence shows young workers recover faster from displacement than older workers (Athey et al., 2024), making early intervention particularly promising.
- Monitoring the **composition** of employment, not just the volume, is essential for timely policy response.

Data & method: Monthly employer declarations (arbetsgivardeklarationer, AGI) from Statistics Sweden, covering *every* employment relationship in Sweden, 2019–June 2025. Outcome: employment positions by age group and occupation. AI exposure measured using the DAIOE index (Engberg et al., 2024). Difference-in-differences with employer×quartile and employer×month fixed effects, following Brynjolfsson et al. (2025). SEs clustered by employer. DAIOE is constructed from pre-2022 work content, fixed across the sample period.

13 March 2026

Code: github.com/Magnus-L/canaries-sweden

Preliminary results. Working paper forthcoming 16 March 2026. Part of AISCAF (WASP-HS) and the AI-Econ Lab at Örebro University and Ratio.

Contact: Lydia Löthman, lydia.lothman@oru.se, 070-9584726 • Magnus Lodefalk, magnus.lodefalk@oru.se, 0722-217340